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An Emerging Theory of Avatar Marketing

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Abstract

Avatars are becoming increasingly popular in contemporary marketing strategies, but their effectiveness for achieving performance outcomes (e.g., purchase likelihood) varies widely in practice. Related academic literature is fragmented, lacking both definitional consistency and conceptual clarity. This article makes three main contributions to avatar theory and managerial practice. First, to address ambiguity with respect to its definition, this study identifies and critically evaluates key conceptual elements of the term *avatar*, offers a definition derived from this analysis, and provides a typology of avatars' design elements. Second, the proposed 2 × 2 avatar taxonomy suggests that the alignment of an avatar's form realism and behavioral realism, across different contingencies, provides a parsimonious explanation for avatar effectiveness. Third, the authors develop an emerging theory of avatar marketing by triangulating insights from fundamental elements of avatars, a synthesis of extant research, and business practices. This framework integrates key theoretical insights, research propositions, and important managerial implications for this expanding area of marketing strategy. Lastly, the authors outline a research program to test the propositions and insights as well as advance future research.

Keywords

avatar, artificial intelligence, behavioral realism, chatbot, form realism, human–computer interaction

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Advances in computer technology have supported the proliferation of virtual characters, broadly known as *avatars*, which we define as digital entities with anthropomorphic appearance, controlled by a human or software, that have an ability to interact. Companies are heavily investing in avatars to engage and serve their customers better, and the use of avatars is predicted to increase by 241% in the travel and hospitality industry and by 187% for consumer goods (Sweezey 2019). In the banking industry, 87% of companies already use some form of an avatar or plan to implement one within two years (Torresin 2019).

Even as the use of avatars continues to rise, their effectiveness varies significantly across firms. For example, Progressive Insurance's avatar, Flo, serves many customers successfully on Facebook Messenger (Briggs 2018), but IKEA withdrew its avatar, Anna, from its online website following its unsatisfactory performance (Brandtzaeg and Følstad 2018). No guidelines exist for effective design or use of avatars (Bradbury 2018), and academic research often lags behind business practices. In addition, it is difficult to integrate extant research on avatars to establish a strong foundation because the literature in this domain is very fragmented and lacks definitional and conceptual precision, as is typical for an emerging research area. With this article, we aim to provide an integrated theoretical framework of avatars by establishing definitional and conceptual clarity, synthetizing academic research and business practices, and offering propositions that include both managerial insights and future research opportunities.

This article aims to make three main contributions to avatar theory and managerial practice. First, because extant research remains ambiguous with respect to defining and creating a taxonomy of avatars, it also remains difficult for researchers to compare empirical results and draw meaningful conclusions across studies. We offer an overview, in which we summarize the various ways the term *avatar* has been defined, identify and

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critically evaluate key conceptual elements of these definitions, and propose a definition on this basis. Then, drawing on this definition, we present a typology of avatar design to isolate elements that academics and managers can leverage to ensure avatars' effectiveness for achieving specific goals (e.g., providing standard vs. personalized solutions). This typology represents an overall organizing framework for thinking about the uses of avatars, making relevant design/implementation decisions, and identifying research gaps.

Second, applying our proposed avatar typology, we synthesize findings from prior literature and business practices to produce a 2×2 taxonomy composed of two dimensions: avatars' form realism and behavioral realism. This taxonomy enables us to generate specific research propositions to advance our understanding of and improve effectiveness of avatars in marketing. The level of alignment between an avatar's form and behavioral realism across different contingencies provides a parsimonious explanation for when an avatar is most effective, which we highlight with examples of successful and failed avatars in business practices. In particular, avatars with high form realism but low behavioral realism, which we call *superficial avatars*, can hinder customer experiences in high-risk transactions (e.g., stock purchases) because the misalignment between an anthropomorphic appearance and low intelligence leads to negative disconfirmations of expectations (e.g., Nordnet's Amelia exhibits realistic human appearance but does not offer good stock tips). In contrast, an *intelligent unrealistic avatar* (i.e., low form realism, high behavioral realism) may produce positive disconfirmations in socially complex interactions that require exchanges of sensitive personal information (e.g., Ellie, an online avatar, helps assess people's depression and posttraumatic stress disorder [PTSD] symptoms); because the avatar's unrealistic human appearance cannot be mistaken for a real human, people are more likely to provide responses that are free from a social desirability bias.

Third, by triangulating insights from the avatar fundamentals and our synthesis of the extant research and business practices, we develop an integrative framework of avatar performance and offer key theoretical insights, research propositions, and managerial implications for this expanding area of marketing strategy. This framework sheds new light on the underlying mechanisms of effective avatar design and implementation strategies, as well as potential contingencies to these effects. In turn, we offer managerial guidelines regarding avatar deployment, specific decision rules, and insights into *how* and *why* specific avatar strategies may be effective. This foundation for a contemporary theory of avatars may spur future research; the identified academic–practice gaps and propositions in particular point to promising research directions.

Avatar Fundamentals

The popularity of avatars is fueled by two macroenvironmental factors. First, advancements in computer/digital technologies (e.g., artificial intelligence [AI]) enabled the development of more complex avatars, and they often appear in three-dimensional (3D) forms, imbued with seemingly distinctive personalities, appearances, and behavioral patterns, and are overall more appealing than the previous, simple versions (Ahn, Fox, and Bailenson 2012; Garnier and Poncin 2013). Second, increase in the use of avatars reflects the growing importance of online service experiences, such as education, gaming, banking, and shopping (Garnier and Poncin 2013; Kim, Chen, and Zhang 2016), which firms want to make as convenient and hassle-free for customers as possible (Kohler et al. 2011). For example, online customers often express frustration when they cannot find relevant information on a website quickly and easily; avatars can effectively and efficiently provide a solution to this issue. Today's customers also expect faster communication from companies, but immediate responses tend to be difficult and expensive for firms to achieve through traditional channels (e.g., face-to-face, telephone) (Kilens 2019). Shopping in the online environment also reduces customers' sense of social interaction and personal consultation, a concern that avatars can help address (Holzwarth, Janiszewski, and Neumann 2006). Finally, recent technology developments enable avatars to offer expanded benefits to firms, in that "avatars capable of having complex and interactive conversations with customers will exponentially increase the amount of data businesses can access. Avatars can potentially know if [customers] were bored or happy in real time and know the exact moment someone became disengaged" (Frank 2019).

Defining Avatars

Although we can readily list the benefits of avatars, no strong consensus exists regarding their precise definition (Table 1). Furthermore, academics have used multiple terms interchangeably to refer to avatars, such as automated shopping assistants (Al-Natour, Benbasat, and Cenfetelli 2011), chatbots (Ho, Hancock, and Miner 2018), virtual customer service agents (Verhagen et al. 2014), embodied conversational agents (Bickmore, Pfeifer, and Jack 2009; Lee and Choi 2017; Schuetzler et al. 2018), or virtual/digital assistants (Chattaraman et al. 2019; Freeman and Beaver 2018). The ambiguity surrounding the definition of avatars makes it difficult for researchers to compare empirical results or draw meaningful conclusions across studies (Nowak and Fox 2018). To advance scientific knowledge, we need a precise definition that clearly delineates the boundaries of the construct. In this section, we review various ways avatars have been defined, identify and critically evaluate some key definitional elements, and offer a new definition derived from this analysis.

Anthropomorphic appearance. One aspect on which there is no consensus when it comes to defining avatars is whether avatars need to have an anthropomorphic appearance (Nowak and Fox 2018). *Anthropomorphism* refers to "the extent to which an image looks human" (Nowak and Rauh 2006, p. 154). In prior academic research, 70% of articles identify an anthropomorphic or humanlike appearance as a necessary condition of

Table 1. Avatar Definitional Elements in Empirical Research.

Illustrative Research	Labels/Aliases	Definitions	Definitional Element				Avatar by Our Definition?
			Digital	Anthropomorphic Appearance	Interactivity	Controlling Entity	
Percentage of element			78%	70%	78%	90%	51%
Ho, Hancock, and Miner (2018)	Chatbot	"Computer programs that can simulate human-human conversation" (p. 712)	✓		✓	Human	No
Holzwarth, Janiszewski, and Neumann (2006)	Avatar	"General graphic representations that are personified by means of computer technology" (p. 20)	✓	✓	✓	Software	Yes
Jin (2009)	Avatar	"Artificial, computer-animated representations of human interlocutors" or "pictorial representations of humans in a chat environment" (p. 234)	✓	✓		Software	No
Kang and Watt (2013)	Avatar	"Digital models of people that either look or behave like the people they represent." (p. 1170)	✓	✓	✓	Human	Yes
Kim, Chen, and Zhang (2016)	Anthropomorphized helper / digital assistant	Entities that "are often imbued with humanlike features and characteristics" (p. 283)	✓		✓	Software	No
Köhler, Rohm, and De Ruyter (2011)	Socialization agent/online agent	"Computer mediated personas that possess the capability to involve customers in rich interactive conversations, rather than discrete, basic exchanges, and that have the ability to apply past interaction content to current interactions" (p. 96)			✓	Software	No
Nunamaker et al. (2011)	Embodied conversational agent	"Virtual, three-dimensional human likenesses that are displayed on computer screens…and interact with people through natural speech" (p. 21)	✓	✓	✓	Software	Yes
Schuetzler et al. (2018)	Conversational agent	"Systems that mimic human-to-human communication using natural language processing, machine learning, and/or artificial intelligence" (p. 94)	✓	✓	✓	Software	Yes
Sivaramakrishnan, Wan, and Tang (2010)	Anthropomorphic information agent	"A humanlike chatbot that acts as an interactive online information provider" (p. 60)	✓		✓	Software	No
Touré-Tillery and McGill (2015)	Anthropomorphized agent (partial human)	"Nonhuman entities that deliver message content across a variety of media (e.g., print, online, television), are typically imbued with various combinations of human characteristics, such as human form (e.g., human-like faces, arms, and legs), and the apparent ability to speak and think" (p. 94)	✓			Software	No
Verhagen et al. (2014)	Virtual customer service agent	"Computer-generated characters that are able to interact with customers and simulate behavior of human company representatives through artificial intelligence" (p. 530)	✓	✓	✓	Software	Yes
Wang et al. (2007)	Virtual character	"Avatar with some type of combination of 4 online social cues: language, human voice, interactivity, and social role" (pp. 143–44)	✓	✓	✓	Software	Yes
This research	**Avatar**	**Digital entities with anthropomorphic appearance, controlled by a human or software, that have an ability to interact**	✓	✓	✓	**Software or human**	

a conceptual definition of an avatar. This element is important because the degree to which an avatar is anthropomorphic provides cues of its social presence (Nass and Moon 2000). Research shows that the more anthropomorphic an avatar is perceived to be, the more credible and competent it seems (Westerman, Tamborini, and Bowman 2015), such that "a person may be represented by a highly accurate and lifelike avatar of a fir tree. Although this avatar is realistic, other users may be less likely to attribute social potential to it—and less likely to communicate with it—because it is not anthropomorphic" (Nowak and Fox 2018, p. 37). Research shows that how anthropomorphic we perceive something to be impacts our expectations of certain behaviors and our willingness to interact; people treat something with a human appearance differently than they do inanimate objects (Fox et al. 2015). For example, Neytiri in the film *Avatar* is not a human, but because she has an anthropomorphic appearance, other characters interact with her the same way they would with a human.

Knowledge about how to deal with other humans generally is learned early in life and is more detailed and readily accessible in people's memory than knowledge about how to interact with inanimate objects (Epley, Waytz, and Cacioppo 2007). According to the computers-as-social-actors (CASA) paradigm (Moon 2000; Nass et al. 1995; Reeves and Nass 1996), people tend to treat computer technology that exhibits humanlike characteristics as a social actor and apply the same social rules to it during interactions, despite being fully aware that they are dealing with a machine (Holzwarth, Janiszewski, and Neumann 2006). The presence of an anthropomorphic appearance triggers people's simplistic social scripts (e.g., politeness, reciprocity), which in turn induce cognitive, affective, and social responses during interactions with technology (Wang et al. 2007). Thus, we regard an anthropomorphic appearance as an important, required element of the conceptual definition of digital avatars, because people interact differently with something they perceive as more "human." This requirement, thus, would exclude inanimate objects and brands as well as voice-only digital assistants that lack an anthropomorphic appearance.

Interactivity. Interactivity refers to "the extent to which individuals perceive that the communication allows them to feel in control as if they can communicate synchronously and reciprocally with the communicator" (Chattaraman et al. 2019, p. 317). In defining interactivity as another critical requirement for digital avatars, we refer specifically to the ability to engage in two-way interactions, which may be verbal (voice) or nonverbal (text, animation). Prior research has established three dimensions of interactivity: (1) the user's active control, or ability to participate and influence communication; (2) bilateral interactions; and (3) synchronicity (Etemad-Sajadi 2016; Liu and Shrum 2002). Approximately 78% of the research we reviewed include interactivity as one of the elements of the conceptual definition of avatars. For example, in defining avatars as "virtual characters that can be used as company representatives in online stores," Liew, Tan, and Ismail (2017, p. 2) allow them to be noninteractive and capable of only one-way conversation, such as welcoming users, introducing the company, or describing available products. Some other researchers similarly do not consider interactivity as a necessary element of digital avatars (e.g., Jin 2009). Yet most researchers focus on interactive avatars and find that they can increase customers' satisfaction with a website or product, credibility, or patronage intentions (Chattaraman, Kwon, and Gilbert 2012; Holzwarth, Janiszewski, and Neumann 2006).

However, designing a truly interactive avatar that can engage in synchronous communication is not an easy task: "Natural language dialogue in chatbots suggests a low threshold for users to access data and services. However, whereas conversational interfaces are truly intuitive when applied to interactions between people, conversations between humans and automated conversational agents are more challenging" (Brandtzaeg and Følstad 2018, p. 41). The interactivity requirement would exclude entities such as "self-avatars" in clothing stores, which do not offer bidirectional communication, as well as any instances of asynchronous content, such as a lecture delivered as a prerecorded video of the instructor, or a standard greeting from a chatbot that cannot offer personalized interactions with each user. However, when there is true bidirectional interactivity, it can satisfy customers' hedonic (e.g., having fun while shopping on a website) and utilitarian (e.g., efficiently finding a solution to a problem on a website) needs (Liew, Tan, and Ismail 2017). Thus, we include it as a requirement for digital avatars.

Controlling entity. Researchers also have different perspectives on the controlling entity, which refers to whether the control over an avatar involves a human operator or an automated computer program (Nowak and Fox 2018). Some researchers make this distinction explicit and refer to anything controlled by technology as an agent or bot, while referring to anything controlled by humans as an avatar (Nowak and Fox 2018). However, in business practices, due to cost considerations, digital avatars appear almost exclusively enabled by AI (e.g., Sophie, the Air New Zealand customer service rep). Yet we have no theoretical reason to limit the conceptual definition of avatars to only those which are enabled by AI, because it seems that consumers want a *perception* of an avatar having some level of intelligence but often cannot tell precisely who or what controls it (Kim and Sundar 2012), as there are typically no solid clues available. According to the modality–agency–interactivity–navigability model and its applications in virtual environments, if agency cues are present in an interface, they influence users' perceptions by prompting their cognitive heuristics about the nature and content of the interaction (Sundar 2008). Users' perceptions and behaviors thus differ if they learn that they are interacting with an AI-backed avatar versus one controlled by a person, reflecting the different heuristics that are evoked by machine versus human counterparts (Fox et al. 2015; Go and Sundar 2019). Thus, "identity cues suggesting that the user is chatting with a human agent or machine agent can trigger human or machine heuristics respectively and

accordingly affect the criteria by which they evaluate the quality of the interaction" (Go and Sundar 2019, p. 305).

In summary, we define *avatars* as digital entities with anthropomorphic appearance, controlled by a human or software, that are able to interact. Among the academic papers we reviewed, about half (51%) include all of these elements in their conceptual definitions of avatars; the inclusion rates for each specific definitional element vary between 70% and 90%.

Typology of Avatar Design

Drawing on this derived definition of avatars, we propose a typology of avatar design. This typology allows academics and managers to isolate elements that make an avatar more or less effective for specific goals, such as providing product information, answering customers' process questions, and so forth. Furthermore, this typology provides an overall organizing framework for thinking about, making design/implementation decisions regarding, and researching avatars.

Different design elements cause avatars to vary in their visual appearances and behaviors during interactions with humans. All of the design elements affect avatars' form realism and behavioral realism. *Form realism* refers to the extent to which the avatar's shape *appears* human, while *behavioral realism* captures the degree to which it *behaves* as a human would in the physical world (Bailenson et al. 2008; Blascovich et al. 2002; Fox et al. 2015). Some researchers argue that behavioral realism is more important than form realism (Blascovich et al. 2002), but both form and behavioral realism are associated with greater avatar usefulness in most contexts (Garau et al. 2003; Kang, Watt, and Ala 2008; Yee, Bailenson, and Rickertsen 2007). Figure 1 provides an overview of all the design elements, with examples, that managers can use to understand and influence the degree of avatars' form and behavioral realism.

Form Realism

Higher form realism may lead users to develop social expectations for their subsequent interactions with avatars (Nowak and Biocca 2003). Managers can impact the degree of form realism of an avatar through the design elements such as spatial dimensions (2D vs. 3D avatars), ability to have movement in the face or body (visually static vs. dynamic avatars), and other characteristics that enhance the perception of "humanness" of avatars, such as signals of gender, race, age, or names.

Spatial dimension. Avatars can be 2D or 3D. In the sample of articles we reviewed, 52% focused on 2D avatars and 48% on 3D avatars. Research indicates that 3D avatars are perceived as more compelling and impactful, relative to 2D versions (Bailenson et al. 2008; Fox et al. 2015; Persky and Blascovich 2007).

Movement. Both technological advancements and customer expectations have prompted the development of more realistic, visually dynamic avatars that can move their bodies and faces (Yun, Deng, and Hiscock 2009). In our sample of reviewed studies, 38% of studies examined static avatars, while 62% considered visually dynamic avatars. For example, Amelia is a visually dynamic avatar that is capable of facial expressions and movement and has been used, with modifications, in various industries such as banking, insurance, and health care (Ipsoft 2018). Visually dynamic avatars with ability for facial expressions can convey emotions, which is especially helpful for customers from different cultural backgrounds: "Avatars with high intensity expression and dynamics allow both the local and global audiences to achieve approximately equal levels in subject identification and emotion perception" (Yun, Deng, and Hiscock 2009, p. 21). Thus, visually dynamic avatars may be more effective for global corporations. Additional evidence indicates that the greater an avatar's ability to exhibit facial expressions, the less perceived human agency is required to exert social influence (Bailenson et al. 2008).

Human characteristics. To enhance form realism, avatars can be designed to include additional "human" elements, such as an identifiable name, gender, race, and age. In the studies we reviewed, out of the aforementioned set of characteristics, gender and age are the most commonly studied ones, followed by name and race. For example, the Air New Zealand virtual customer service avatar is female, named Sophie. Research shows that characteristics such as gender can increase the effectiveness of avatars (Nass and Yen 2010).

Behavioral Realism

Avatars' behavioral realism can facilitate more natural interactions with users (Blascovich et al. 2002), and managers can manipulate the degree of avatars' behavioral realism using design elements associated with avatars' interactivity and controlling entity. Specifically, relevant design elements that managers can use to impact avatar interactivity are communication modality (avatars' ability to communicate verbally, nonverbally, or through a combination of both), response type (whether avatars' responses are scripted or natural), and presence of social content (whether avatars can engage in interactions about social and personal matters in addition to task-oriented communications). In terms of controlling entity, avatars can be controlled by a computer program or algorithm or a human, with the latter, predictably, increasing avatars' behavioral realism.

Communication modality. Avatars differ in the modalities of communication they use. Nonverbal avatar communication can be represented by text (speech-to-text avatars), gestures, or facial expressions; verbal avatars communicate via speech; and both nonverbal and verbal avatars can communicate using a combination of these modalities. The latter category would be the highest in behavioral realism from a communication modality perspective. Research investigating the communication modality of avatars has focused primarily on avatars that are capable of both verbal and nonverbal interactions, accounting

Managerial Decision	Definitional Element	Design Element	Design Element Level	Definition	Avatar Example[a]	Study Count / %
Degree of **Form Realism**	Anthropomorphic appearance	Spatial dimension	2D	Avatar is displayed as a two-dimensional visual representation	ING bank's avatar Inge has a 2D, cartoonlike visual representation	26 / 52%
			3D	Avatar is displayed as a three-dimensional visual representation	Replika, developed to be a personal AI friend, is a 3D avatar	24 / 48%
		Movement	Static	Avatar without any movement; a static graphic	ABIe, an avatar for Allstate Insurance for business, does not have capability for any movement	19 / 38%
			Dynamic	Avatar is able to have visual movement of body and/or face	Amelia, used by many companies, exhibits facial expressions and movement	31 / 62%
		Human characteristics	Gender	Avatar has a specific gender	Olivia is a symptom-checking avatar for the UK's National Health Service and looks female	47 / 94%
			Race	Avatar is of a specific race	Swedish bank SEB avatar Aida looks White	4 / 8%
			Age	Avatar has an identifiable age range	Eviebot is an AI companion that appears to be in the range of 20-40 years of age	44 / 88%
			Name	Avatar has a name	The Air New Zealand customer service avatar is named Sophie	23 / 46%
Degree of **Behavioral Realism**	Interactivity	Communication modality	Nonverbal	Avatar communicates through text (speech-to-text avatars), and/or has gestures and/or facial expressions	Friendship avatar Mitsuku supports customers' need for connectedness through text communication	13 / 26%
			Verbal	Avatar communicates via speech	GMF's avatar Prosper provides services to customers regarding life insurance policies via interactive speech.	3 / 6%
			Nonverbal & verbal	Avatar has both non-verbal and verbal communication abilities	Microsoft's Xiaoice can communicate via text and voice and can also interpret visuals, such as photos	34 / 68%
		Response type	Scripted	Avatar can provide only a response from a stored set of preexisting and predetermined responses	HSBC's avatar Amy covers only some of the bank's products with predetermined responses	30 / 60%
			Natural	Avatar can have "relatively free-flowing conversation, using accepted vocabulary and grammar, and with the ability to track the context of the conversation and make appropriate responses"[b]	Skincare company SK-II's YUMI understands customers using their own words and responds in an organic, conversational manner	20 / 40%
		Social content	Yes	Avatar can communicate about social/personal matters, in addition to task-oriented communication	Microsoft's Xiaoice communicates about anything that is of interest to the user	17 / 34%
			No	Avatar is unable to communicate about social/personal (to the user) matters and is only able to provide task-oriented communication	Molly, a virtual nurse, collects health information from patients (e.g., postoperative, chronic conditions), suggests follow-up behaviors, and communicates the information to the treating physician. The interaction is limited to health content.	33 / 66%
	Controlling entity	Software-controlled		Avatar is controlled by computer software or algorithm	Progressive Insurance avatar Flo uses machine learning and a cloud-based API to obtain data from social media responses and training data to provide answers to customers	43 / 86%
		Human-controlled		Avatar is controlled by a human in real time	Resident avatars in Second Life are controlled by participants who can interact with one another in real time	7 / 14%

Figure 1. Typology of avatar design.

[a]Although these specific avatar examples are used to illustrate a specific characteristic, they also satisfy all of the other requirements of our conceptual definition of avatars.

[b]Burden and Savin-Baden 2019, p. 9.

Notes: For live links to the URLs noted herein, see the Supplemental Material link.

for 68% of reviewed articles, followed by research on nonverbal avatars in 26% of articles, with verbal avatars attracting the least academic attention (6%).

In addition, managers might increase behavioral realism by enabling avatars to recognize the nonverbal behaviors of users, such as their facial expressions, prompting more appropriate responses. For example, Microsoft's XiaoIce can interpret users' photos and make relevant inferences and comments (Dormehl 2018). However, even with significant advances in AI, creating an avatar capable of correctly identifying and responding to users' various emotions and contexts remains a challenge because "large, interpersonal variability exists in how people express emotions. Humans also have diverse preferences for how an agent [avatar] responds to them" (McDuff and Czerwinski 2018, p. 76).

Response type. Managers can design avatars with an ability to converse in a way that feels natural to users. Previous research has mostly (60% of articles) focused on avatars that are capable of merely selecting a response from a set of preexisting, predetermined, scripted responses. For example, HSBC's virtual assistant, Amy, currently is able to select and provide users with only predetermined responses about a limited number of the bank's products. Avatars with a capability for natural responses instead can have a "relatively free-flowing conversation, using accepted vocabulary and grammar, and with the ability to track the context of the conversation and make appropriate responses" (Burden and Savin-Baden 2019, p. 9). For example, the skincare company SK-II's YUMI understands users expressing themselves in their own words and responds in an organic, conversational manner. The ability to have a conversation that feels natural is also highly correlated with perceived agency type; avatars controlled by humans would have a natural response, whereas software-controlled avatars tend to rely on scripted responses.

Social content. Another design element that can increase avatars' interactivity is their ability to provide some social content during interactions with users, as opposed to purely task-oriented communication (e.g., providing product information). For example, Microsoft's XiaoIce is an AI assistant that also attempts to function like a friend, checking on users after a relationship breakup or asking about the physical recovery of a user who posted a photo of a bruised leg. Since its launch in China in 2014, XiaoIce has gained great popularity, due to its emotional intelligence: "The real key takeaway is that we've focused on emotional intelligence . . . . We call this an empathetic computing framework, [designed to] have conversations with humans naturally, which can build a social and emotional connection. It's a good friend. As a result, they can better participate and help out in human society" (Dormehl 2018). In the research we reviewed, 34% of avatars could offer some social content during their interactions.

Controlling entity. Research shows that consumers interacting with an avatar that they perceive to be controlled by a human behave differently from customers who believe the avatar is controlled by software. According to a meta-analysis, avatars controlled by humans elicit more presence and a stronger social influence than do computer-controlled avatars (Fox et al. 2015). Therefore, for firms that rely on software-controlled avatars (the majority of avatars in practice), reinforcing "human" elements can be very effective. They can leverage the design elements we have discussed, such as more natural speech programmed for software-controlled avatars, nonverbal behaviors and emotions, and the ability to provide social content. However, perceptions of human agency might not be desirable in all settings, as research shows that people perform worse on certain tasks when they recognize that they are interacting with a human-controlled avatar rather than software, due to social inhibition, social desirability bias, and perceptions of reduced autonomy (Kim, Chen, and Zhang 2016; Yokotani, Takagi, and Wakashima 2018). For example, avatars have become popular in health care, and when it comes to disclosing sensitive information such as drinking habits, users are more comfortable revealing information if they perceive less human agency (Schuetzler et al. 2018).

Avatar Academic Research and Business Practices

In the proposed typology, all of the design elements serve to increase or decrease the form and behavioral realism of avatars. Most prior research investigates only a narrow set of design elements in piecemeal fashion, independent of other elements, or at an aggregate level, without considering the granularity or interactions of specific elements. In the following sections, we synthesize academic literature and business practices related to avatars to provide insights and identify research gaps. We also derive propositions to help advance theory, inform business practices, and guide future research.

Academic Literature

Because research on avatars is diverse and cross-disciplinary in nature, our goal is to provide a representative rather than exhaustive literature review, covering a variety of research disciplines and empirical settings. Using "avatar" and related terms[1] as keywords, we searched article titles and abstracts in electronic databases (Academic Search Complete, Business Source Complete, Science Direct, and Google Scholar) to find studies published during the 1990–2020 period. To identify research involving avatars that are consistent with our definition, we first excluded articles that did not provide detailed information about avatar definitional elements or scope. In addition, we excluded research on topics that fall outside our

[1] Related terms included animated agent, anthropomorphic agent, artificial agent, chatbot, conversational agent, digital assistant, electronic shopping agent, embodied agent, embodied virtual agent, nonhuman agent, spokes-avatar, spokes-character, virtual agent, virtual assistant, and virtual character.

definitional boundary (e.g., consumers' self-avatars). Purely technical articles (e.g., programming of avatar) and studies that do not address real-time consumer–avatar interactions were removed as well. To strike a good balance between research diversity and quality, we sampled only reputable journals across various disciplines (i.e., impact factor of at least 3).[2] In total, we compiled 98 empirical research articles (the full list of reviewed research is available in the Web Appendix). Table 2 provides a summary of select illustrative research.

Form realism. The overarching theoretical framework that guides empirical studies of human–avatar interactions is social response theory, which is sometimes referred to as the CASA paradigm (Moon 2000; Nass et al. 1995; Reeves and Nass 1996). It suggests that anthropomorphic characteristics of avatars elicit consumers' socialness perceptions, often in an automatic, spontaneous, mindless process that induces varying degrees of cognitive, affective, and social responses to avatars (Al-Natour, Benbasat, and Cenfetelli 2011; Holzwarth, Janiszewski, and Neumann 2006; Verhagen et al. 2014; Wang et al. 2007). Although the theory suggests that an anthropomorphic appearance positively affects customer outcomes, empirical results indicate some mixed effects. In various situations, lower or higher levels of form realism appeared more effective, but in other cases, no differences emerged. For example, visually static, cartoonish avatars with very low form realism increased satisfaction with a retailer, attitude toward products, and purchase intentions in some studies (Etemad-Sajadi 2014; Holzwarth, Janiszewski, and Neumann 2006). However, Qiu and Benbasat (2009) find that avatars with more realistic, humanlike appearance increased customers' perceptions of social presence, leading to higher usage intentions. Verhagen et al. (2014) found no significant differences between avatars that are low or high in form realism in terms of service satisfaction. Similarly, Schuetzler et al. (2018) reported that a more anthropomorphic appearance of an avatar had no effect on participants' disclosure of information about sensitive topics, such as drinking behaviors.

Two factors may help explain these inconsistent effects. First, prior studies have not investigated all of the design elements identified in our typology that help establish avatars' form realism (e.g., visually static vs. dynamic avatars, avatars' age, gender) or how these underlying elements might induce specific effects. By focusing on only a subset of visual characteristics and, thus, failing to account for the totality of the elements that establish form realism of an avatar, these studies may have produced biased estimates. Second, we propose that avatars' form and behavior must be considered simultaneously, because form realism is meaningful only in the context of behavioral realism (Bailenson et al. 2008); however, few studies have done so.

Behavioral realism. Extant research consistently shows positive effects of greater behavioral realism. For example, Wang et al. (2007) report that an avatar's scripted text or spoken communications can enhance customers' hedonic and utilitarian benefits when shopping online, as well as increasing their patronage intentions. Similarly, Lee and Choi (2017) find that when users interact with an avatar that has a high degree of behavioral realism, trust between the parties is higher. Other studies offer similar results, noting that avatars' behaviors, such as decision-making style (Al-Natour, Benbasat, and Cenfetelli 2011) and socially oriented communication (Verhagen et al. 2014), significantly affect avatar trustworthiness and the overall customer experience (Brave, Nass, and Hutchinson 2005; Chattaraman et al. 2019).

Advancements in AI technology have enabled avatars to exhibit higher levels of cognitive and emotional intelligence. For example, they can engage in autonomous conversations by analyzing and responding to customers' requests in real time, thereby significantly increasing customers' trust in them (McDuff and Czerwinski 2018). Using avatar interviewers equipped with video, audio sensors, and advanced analytical software, Nunamaker et al. (2011) demonstrate how intelligent avatars can detect, interpret, and respond to human interviewees' emotions, cognitive effort, and potential deceptions. Von der Pütten et al. (2010) examine the effects of an avatar that can collect and analyze a human's voice and upper-body movement and coordinate its own responses accordingly. Results show that the avatar's intelligent behavior led to positive evaluations of the avatar, regardless of whether it was controlled by a human or a software.

Even in light of these consistent findings related to avatar behavioral realism, some important research issues remain unresolved. First, few studies have investigated the underlying behavioral realism elements identified in our typology of avatar design (e.g., communication modality, social content, response type) to determine which are most critical or how they might interact with other form or behavioral realism elements. For example, Bickmore, Pfeifer, and Jack (2009) found that an avatar nurse incorporating *social content* in its scripted conversations produced better patient experiences, but Schuetzler et al. (2018) reported that a scripted, *task-focused* avatar interviewer elicits more socially biased responses. Second, a few studies revealed some unexpected negative effects of behavioral realism (e.g., Schuetzler et al. 2018), but research has yet to identify the conditions in which detrimental effects are more likely or design strategies to address them.

Integrated perspective on form and behavioral realism. Our review of extant literature thus reveals a key limitation: lack of consideration of the alignment between form and behavioral realism of avatars. If the levels of form and behavioral realism are mismatched, the consequences for avatars' effectiveness may be profound and can help explain inconsistent past

[2] The academic disciplines included were (1) marketing (e.g., *Journal of Marketing*), (2) computer science (e.g., *Computers in Human Behavior*), (3) information systems (e.g., *Journal of Management Information Systems*), (4) communications (e.g., *Human Communication Research*), (5) education (e.g., *Computers & Education*), (6) health care (e.g., *Journal of Medical Internet Research*), and (7) general business (e.g., *Journal of Business Research*).

Table 2. Avatars in Empirical Research.

Illustrative Research	Original Label	Context	Theoretical Perspective	Mediator Variables	Moderator Variables	Key Findings
Al-Natour, Benbasat, and Cenfetelli (2011)	Automated shopping assistant	Online shopping for a laptop computer	Computers as social actors (CASA) framework; similarity-attraction hypothesis	Perceived decision process similarity	None	Perceived decision process similarity mediates the effect of perceived personality similarity on several beliefs (enjoyment, social presence, trust, ease of use, and usefulness).
Bickmore et al. (2016)	Embodied conversational agent	Cancer patients identifying and learning about clinical trials on the internet	Not specified	None	None	Patients were more satisfied with the conversational agent compared to the conventional web form-based interface, and patients with low health literacy had a higher success rate in finding relevant trials.
Brave, Nass, and Hutchinson (2005)	Embodied computer agent	Casino-style blackjack game	CASA framework	None	None	Empathic emotion of the agent leads to greater user-rated likeability, trustworthiness, perceived caring, and perceived support.
Chattaraman et al. (2019)	Digital assistant	Online purchase of athletic shoes by older consumers	Social response theory	Trust in online store; information overload; perceived self-efficacy; ease of use; usefulness	Internet competency	Users' internet competency interacts with the digital assistant's conversational style (social- vs. task-oriented) in affecting social, functional, and behavioral intention outcomes.
Chattaraman, Kwon, and Gilbert (2012)	Virtual agent	Online purchase of apparel by older consumers	Social response theory; CASA framework	Perceived social support; trust in online store; perceived risk	None	Virtual agents can increase older users' patronage intentions by enhancing perceived social support and trust in the online store while reducing perceived risk.
Derrick and Ligon (2014)	Embodied conversational agent	An avatar-based screening checkpoint experiment	CASA framework	None	Gender	Self-promoting agents are perceived as more powerful, more trustworthy, and more expert, whereas ingratiating agents are perceived as more attractive. Ingratiation impression management techniques are viewed less (more) favorably by females (males) than self-promotion techniques.
D'Mello, Graesser, and King (2010)	AutoTutor	Computer literacy learning with a fully automated computer tutor	Social agency theory	None	None	Students who interacted with the AutoTutor through a spoken dialogue used more cognitive resources and completed more problems than students who had to type.
Go and Sundar (2019)	Chatbot	Online digital camera purchase	CASA framework	Social presence; homophily	Anthropomorphic visual cues; agency	Chatbot's message interactivity has positive effects on customers' attitude toward the website and return intention mediated by perceived social presence and homophily; anthropomorphic visual cues and agency moderate these effects.

(continued)

Table 2. (continued)

Illustrative Research	Original Label	Context	Theoretical Perspective	Mediator Variables	Moderator Variables	Key Findings
Holzwarth, Janiszewski, and Neumann (2006)	Avatar	Online purchase of shoes that are customizable via online consultation	Social response theory	Entertainment value; information value; likeability of avatar; credibility of avatar	Product involvement	Use of an avatar sales agent increases satisfaction with the retailer, attitude toward the product, and purchase intentions, mediated by perceived entertainment and information value; an attractive avatar is more effective at moderate levels of product involvement, mediated by likeability of the avatar, whereas an expert avatar is more effective at high levels of product involvement, mediated by credibility of the avatar.
Keeling, McGoldrick, and Beatty (2010)	Avatar	Online experiments of retail websites selling books/CDs and travel insurance	CASA framework	Trust perception	Goods/services high in credence vs. search qualities	Avatars' social orientation and task orientation increase customers' trust perception, which subsequently has a positive effect on patronage intention. Effects of task- (social-) oriented communications are stronger for search (credence) goods/services.
Kim, Hong, and Cho (2007)	Intelligent conversational agent	Online electronic product (e.g., cellular phone) information search	Not specified	None	None	Agents capable of the probabilistic inference and the semantic inference show superior performance in providing suitable responses to user inquiries with only a few interactions.
Lee and Choi (2017)	Conversational agent	Interactive movie recommendation system	CASA framework; media equation theory; uncertainty reduction theory	Intimacy; trust; interactional enjoyment	None	Self-disclosure and reciprocity of the conversational agent have positive impacts on user satisfaction and intentions to use, mediated by intimacy, trust, and interactional enjoyment.
Mimoun and Poncin (2015)	Embodied conversational agent	Furniture purchase with Anna on IKEA's website	Technology acceptance model	Utilitarian value; hedonic value	None	Anna increased consumers' satisfaction and behavioral intentions through utilitarian and hedonic value.
Nunamaker et al. (2011)	Embodied conversational agent	Automated kiosk-based interviews	Not specified	None	None	Male-embodied agents are perceived as more powerful, more trustworthy, and more expert than female ones; however, the latter are more likeable. Avatars with neutral expressions are perceived as more powerful, whereas smiling avatars are more likeable.
Von der Pütten et al. (2010)	Embodied conversational agent	Interactions involving personal questions	Threshold model of social influence; Ethopoeia concept	None	None	Beliefs about whether a participant is interacting with a human-controlled or a computer-controlled agent lead to almost no differences in the evaluation of the virtual character or its behavioral reactions; higher behavioral realism affected both.

(continued)

Table 2. (continued)

Illustrative Research	Original Label	Context	Theoretical Perspective	Mediator Variables	Moderator Variables	Key Findings
Qiu and Benbasat (2009)	Anthropomorphic interface agent	Online recommendation system for complex and attribute-intensive digital cameras	CASA framework; social agency theory	Social presence	None	Humanoid appearance and human voice-based communication of avatars significantly increase participants' perceived social presence, which has a positive effect on trust, perceived usefulness, perceived enjoyment, and the decision to use the avatar as a decision aid.
Schuetzler et al. (2018)	Conversational agent	Responses to sensitive questions to a person vs. a conversational agent vs. online survey	Self-disclosure; social desirability; social presence theories	None	None	Conversational agents with better conversational abilities prompt more socially desirable responses from participants, with no significant effect found for embodiment.
Verhagen et al. (2014)	Virtual customer service agent	Inquiries about online mobile phone service	Social response; implicit personality; primitive emotional contagion; social interaction theories	Social presence; personalization	Communication style (socially vs. task-oriented); anthropomorphism	Friendliness and expertise have positive effects on participants' service encounter satisfaction mediated by social presence and personalization; the effect of friendliness on personalization is stronger for socially oriented agents than for task-oriented agents, as is the effect of expertise on social presence.
Wang et al. (2007)	Virtual character	Online travel information service	Social response theory; stimulus-organism-response framework; cognitive mediation theory	Arousal; pleasure; flow; hedonic and utilitarian values	Product involvement	Social cues from interacting with the avatar increase perceptions of website socialness, feelings of arousal, pleasure, and flow, leading to greater hedonic and utilitarian values, which then increase patronage intentions. The effect of arousal on pleasure is stronger when product involvement is high; the influence of arousal on hedonic value is stronger for women. Flow does not lead to pleasure for older consumers, and pleasure has a much weaker impact on utilitarian value for those consumers.
Yokotani, Takagi, and Wakashima (2018)	Virtual agent	Mental health interviews	Threshold model of social influence	None	None	Participants revealed more sex-related symptoms to the virtual agent than to a human expert, whereas they disclosed mood and anxiety symptoms more often to the human expert than to the virtual agent.

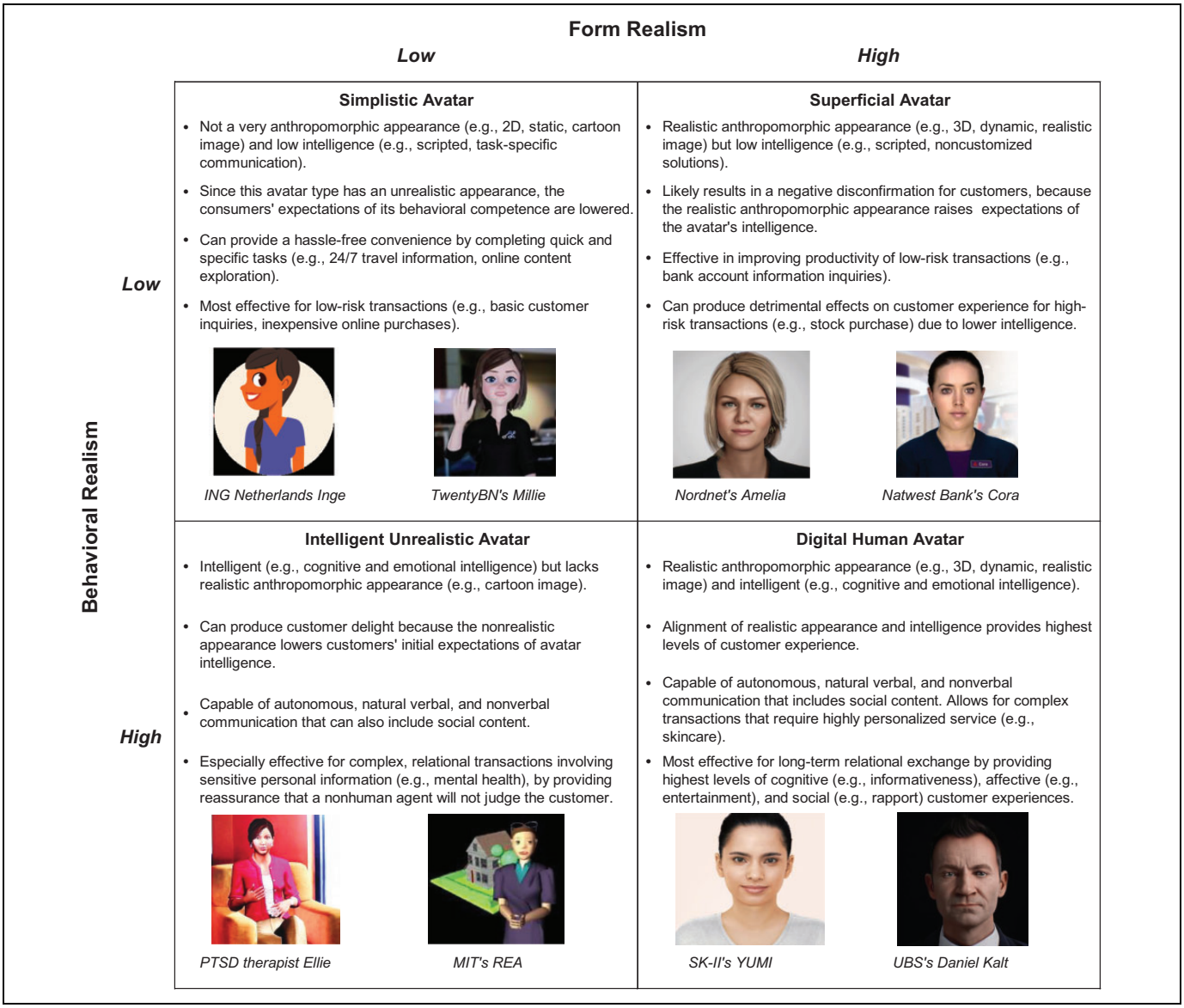


Figure 2. Form realism versus behavioral realism taxonomy.

findings. Yet some misaligned avatars (e.g., the REA avatar is high in behavioral realism but low in form realism) seem equally as effective as well-aligned avatars (e.g., SK-II's skincare advisor YUMI is very high in both behavioral and form realism). However, other misaligned avatars have failed (e.g., Nordnet's Amelia, with high form realism but low behavioral realism). A systematic analysis of avatar effectiveness thus seems warranted and requires identifying different categories of avatars along the form and behavioral realism dimensions as a first step.

We suggest that avatars can be parsimoniously grouped into a 2 × 2 taxonomy, according to their form and behavioral realism (Figure 2). This taxonomy provides a foundation for predicting the success or failure of avatars in business practices and can inform avatar design strategies. We identify four distinct categories of avatars: simplistic, superficial, intelligent unrealistic, and digital human avatars. A simplistic avatar has an unrealistic human appearance (e.g., 2D, visually static, cartoonish image) and engages in low intelligence behaviors (e.g., scripted, only task-specific communication). For example, in the Netherlands, ING Bank uses a 2D, cartoonish-looking avatar, Inge, to provide responses to simple customer inquiries with a set of predetermined answers. In contrast, a superficial avatar has a realistic anthropomorphic appearance (e.g., 3D, visually dynamic, photorealistic image), such as Natwest Bank's Cora, but low behavioral realism, in that it is only able to offer preprogrammed answers to specific questions. An intelligent unrealistic avatar (e.g., REA) is characterized by humanlike cognitive and emotional intelligence but exhibits an unrealistic (e.g., cartoonish) human image. These avatars can engage customers in real-time, complex transactions without being mistaken for real human agents. Finally, a digital

human avatar such as SK-II's YUMI is the most advanced category of avatars, characterized by both a highly realistic anthropomorphic appearance and humanlike cognitive and emotional intelligence, and is designed to provide the highest degree of realism during interactions with human users.

Insights and propositions derived from academic literature. To advance extant literature, we propose the need to consider the interrelationship of form and behavioral realism. Visual information (i.e., what an avatar looks like) often gets processed automatically and almost immediately, requiring minimum cognitive resources (McGloin et al. 2009). This visual appearance then becomes the basis for *probabilistic consistency inferences*, where consumers form an expectation of some unknown attribute, based on a known attribute with which it is believed to be correlated (Dick, Chakravarti, and Biehal 1990). For example, consumers often make inferences about an unfamiliar brand's quality by using price as a signal of quality, in the belief that the two are correlated. Similarly, when an avatar looks more like a human, consumers may expect it to also behave like a human. Thus, the visual characteristics of an avatar may influence consumers' judgments of its behavioral competence, even before an interaction takes place (Nowak and Biocca 2003). A more realistic anthropomorphic appearance suggests a higher level of behavioral realism, leading to a greater expectation that the avatar will behave like a real human might.

> **Proposition 1:** As the form realism of an avatar increases, so do customers' expectations for its behavioral realism.

Customers will use an avatar's form realism as a frame of reference for forming initial expectations about its behavioral realism. The expected level of behavioral realism will then serve as a benchmark, against which consumers will form comparative judgments of their subsequent experience. According to expectation confirmation theory (Oliver 1980), when the actual outcome is worse than expected, consumers experience a *negative* disconfirmation, leading to decreased overall satisfaction. A better-than-expected outcome instead results in a *positive* disconfirmation, which increases customers' overall satisfaction (Evangelidis and Van Osselaer 2018; White and Schneider 2000).

Consistent with this theory, if an avatar's behavioral realism exceeds the consumer's initial expectations, which were based on the avatar's form realism, a positive disconfirmation likely occurs, and the consumer should perceive the avatar as more credible and attractive, as well as feel increased trust or confidence in it (Afifi and Burgoon 2000). Ellie, an avatar that assesses depression and PTSD symptoms in veterans, serves as a good illustration of a positive disconfirmation: her cartoonish appearance paired with highly intelligent, humanlike behavior has proven highly effective with vulnerable individuals (Robinson 2015). Conversely, if the avatar's behavioral competency falls short of the expectations that users formed on the basis of the avatar's form realism, it may lead to a negative disconfirmation and dampen customers' satisfaction (McGloin et al. 2009). Nordnet's Amelia exhibited minimal competence in providing customized advice for high-risk transactions (e.g., stock purchase), which proved disappointing. Amelia's realistic anthropomorphic appearance might have led customers to develop high behavioral expectations, which Amelia could not deliver, giving rise to a negative disconfirmation. Therefore, we expect asymmetric effects of misaligned avatar form and behavioral realism.

> **Proposition 2:** Differences between the avatar's form and behavioral realism have asymmetric effects, such that customers experience *positive (negative)* disconfirmation when an avatar's behavioral realism is *greater (less)* than its form realism.

Mediation mechanisms of avatar effects on performance. Integration of evidence across multiple research streams suggests that avatars affect performance outcomes (e.g., customers' likelihood to purchase a product) indirectly through customers' cognitive, affective, and social responses, depending on the context (Holzwarth, Janiszewski, and Neumann 2006; Lee and Choi 2017). Customers form cognitive responses to avatars according to the avatars' informativeness or competence in helping them make well-informed decisions (Holzwarth, Janiszewski, and Neumann 2006; Wang et al. 2007). Cognitive trust, or willingness to rely on another entity's help to achieve goals in uncertain situations, is another key dimension of customers' cognitive response (Keeling, McGoldrick, and Beatty 2010; Martin, Borah, and Palmatier 2017).

Interactions with avatars can also evoke affective responses in customers, such as by providing them with unique entertainment experiences (Holzwarth, Janiszewski, and Neumann 2006). Avatars can deliver pleasurable experiences independent of their ability to facilitate a specific functional goal, such as a shopping task, by offering entertainment and emotional value during the shopping process (Wang et al. 2007). Human–avatar interactions are also social in nature. Avatars can enhance customers' perceived social presence (i.e., the feeling of being with another person) and create feelings of human contact or connection (Chattaraman et al. 2019; Qiu and Benbasat 2009). Moreover, the use of an avatar can provide a sense of personalization, so customers receive information that appears tailor-made to their specific needs (Verhagen et al. 2014; Wang et al. 2007). The CASA framework argues that avatars can also induce feelings of reciprocity in human–computer interactions, which can strengthen perceived rapport with the avatar and enhance the users' social experience (Chattaraman et al. 2019; Lee and Choi 2017).

When the levels of an avatar's form realism and behavioral realism are aligned, customers' behavioral expectations tend to be confirmed. This simple confirmation, together with high initial behavioral expectations induced by form realism, may have a strong, positive, additive effect on performance outcomes, such as customers' purchase likelihood (Oliver 1980; Stayman, Alden, and Smith 1992). We expect that avatars that are aligned in their form and behavioral realism can affect

customer performance outcomes through all three types of mediating responses: cognitive, affective, and social. However, when the levels of form and behavioral realism are misaligned, the outcomes might be mediated through different responses. Consider the misalignment that occurs when form realism exceeds behavioral realism. In this situation, customers may find the avatar to be especially entertaining, because its realistic anthropomorphic appearance and characteristics can also serve as hedonic elements that can increase perceived entertainment, which often is intrinsically enjoyable in its own right, regardless of performance outcomes (Davis, Bagozzi, and Warshaw 1992; Xu, Abdinnour, and Chaparro 2017). To the extent that perceived enjoyment creates a pleasant mood, the hedonic aspect of high form realism can improve performance outcomes such as impulsive online purchases (Parboteeah, Valacich, and Wells 2009). Yet customers also might experience negative disconfirmation, stemming from the disappointment with an avatar's cognitive and social capabilities, resulting in weakened customer performance outcomes.

Alternatively, when misalignment arises because the avatar's behavioral realism exceeds its form realism, the positive effects on customer outcomes might be mediated primarily by cognitive and social responses. Although typically this avatar's lower form realism is unlikely to provide much entertainment for the customer,[3] the positive disconfirmation of avatar's behavioral competence may significantly boost customers' confidence in the avatar's overall ability to provide valid information, offer personalized service, or build customer rapport.

> **Proposition 3:** When an avatar's form realism exceeds its behavioral realism, it has a positive effect on performance outcomes (e.g., purchase likelihood), through customers' (a) affective responses, but a negative effect on performance outcomes through customers' (b) cognitive and (c) social responses.
>
> **Proposition 4:** When an avatar's behavioral realism exceeds its form realism, it has a positive effect on performance outcomes (e.g., purchase likelihood) through customers' (a) cognitive and (b) social responses.

Business Practices

As we noted previously, many companies are adopting avatars to humanize their brands with a scalable personalized human touch, but managers lack guidance for how to design these avatars to ensure their effectiveness (Bradbury 2018; Wooler 2019). In this section, we use business examples to illustrate the avatar categories from our taxonomy (Figure 2) and thereby clarify which factors and conditions make them effective. We also generate theoretical insights and managerial implications.

Simplistic avatar. An obvious benefit of using avatars is the firm's improved efficiency and scalable customer service. For example, ING Bank's cartoonish avatar, Marie, answers common debit and credit card questions with preprogrammed information and solutions (www.ing.com). In the Los Angeles Superior Court, an animated cartoon avatar, Gina, which speaks multiple languages, successfully handles 1.2 million new traffic citations a year (Llop 2016). A start-up company called TwentyBN has introduced an animated cartoon sales avatar, Millie, which can understand and answer simple questions while presenting various products (Kahn 2018) and seems to be especially effective in promoting low-ticket items, such as sunglasses. Simplistic avatars thus seem most effective in providing hassle-free, convenient options for completing quick, specific tasks (e.g., information inquiries), especially when relatively little risk is involved (e.g., inexpensive online purchases).

Superficial avatar. The use of superficial avatars in various industries shows more mixed results. Among the successes, HSBC Hong Kong's Amy, a photorealistic avatar that handles routine customer inquiries similar to ING's Marie, was well-received by customers (Torresin 2019). A very realistic-looking 3D avatar, Cora of Natwest Bank in the United Kingdom, can answer 200 basic questions, such as how to open an account or complete a mortgage application (Peddie 2018). In the insurance industry, Lemonade Insurance's avatar Maya and Progressive's Flo, both very humanlike, are programmed to provide category-specific information and handle simple transactions, such as onboarding customers and giving online quotes (Briggs 2018; Phaneuf 2020). However, other superficial avatars have been less effective. The Swedish bank Nordnet had to discontinue its realistic-looking avatar Amelia, presumably due to her failure to provide intelligent stock purchasing advice. At IKEA, the decision to eliminate its avatar Anna stemmed from a recognition that her realistic anthropomorphic appearance led to complex customer questions, which required responses beyond the predetermined set available in the avatar's programming (Brandtzaeg and Følstad 2018; Scott 2008). Overall, superficial avatars can entertain customers while enhancing efficiency in low-risk transactions (e.g., bank account information inquiries), but they also can produce detrimental effects for customers seeking high-risk or complex transactions (e.g., financial investments) because these avatars lack the level of intelligence that their realistic anthropomorphic appearance leads users to expect.

Intelligent unrealistic avatar. This type of avatar is relatively rare but generally successful. For example, the REA avatar has been effective in providing virtual showings of homes for sale; Ellie, an avatar therapist, has been useful in detecting PTSD and depression symptoms in military veterans. With its humanlike intelligence, Ellie can engage in context-appropriate, autonomous conversations and build rapport with subtle, supportive, and sympathetic gestures when listening to a veteran's sensitive story. In turn, veterans disclose significantly more PTSD

[3] We acknowledge there may be exceptions in which a very "cute" avatar, although low in form realism, might provide strong hedonic value. We thank an anonymous reviewer for this insight.

symptoms to her than to a human therapist (Gonzalez 2017). Thus, intelligent unrealistic avatars seem especially effective for complex relational transactions involving sensitive personal information (e.g., finances, health) as they can provide a sense of nonjudgment because customers recognize that these avatars are not human but are still competent in their tasks.

Digital human avatar. With advanced digital and computing technologies, pioneer avatar companies such as Soul Machines are breaking new ground for digital human avatars in marketing applications (www.soulmachines.com). For example, skincare brand SK-II uses an incredibly realistic looking and behaving avatar, YUMI, whose AI-powered digital brain enables advanced cognitive and emotional intelligence. In addition, YUMI can recognize users' gestures and features, such as eye color; communicate via speech or text; and deliver customized tips with credible, highly personalized beauty advice (Brunsman 2019). Another digital human avatar, modeled after Daniel Kalt, the chief economist of UBS investment bank, can forecast financial data and present investment advice to high-wealth customers (www.nanalyze.com). Overall, digital human avatars seem most effective for building long-term customer relationships in contexts that feature substantial complexity or risk (e.g., financial investments), where users prioritize realistic, trustworthy, and personalized service.

Insights and Propositions Derived from Business Practices

Observations from business practices suggest that avatars' effectiveness may be highly contingent on the level of perceived uncertainty users experience during their interactions with avatars. This uncertainty might arise from contextual factors, such as functional risk, financial risk, or price (De Haan et al. 2018). *Functional risk* refers to the concern that the product/service may fall short of performance expectations. *Financial risk* refers to a possible loss of money, independent of purchase price, due to a poor decision (e.g., stock performance) (De Haan et al. 2018). In addition, as the purchase *price* increases, the need for information about the quality of products or services also becomes more important to manage perceived risk (Wu and Wang 2005). Overall, we predict that when customers feel greater uncertainty, they develop heightened expectations that an avatar that offers a realistic anthropomorphic appearance will also have a comparable level of behavioral realism, because they rely on the avatar's informativeness and ability to provide personalized advice to reduce the perceived risks associated with the purchase.

> **Proposition 5:** The positive effect of an avatar's form realism on customers' behavioral realism expectations is stronger when (a) functional risk is higher, (b) financial risk is higher, and (c) the product is more expensive.

We previously posited that when form realism exceeds behavioral realism, the avatar can have both positive (via customer's affective responses) and negative (via customer's cognitive and social responses) effects on performance outcomes. These mediated effects also might be moderated by perceived uncertainty. When perceived uncertainty is high, the avatar's entertainment value may become less salient to the consumer because an entertaining avatar cannot overcome the perceived risks associated with the purchase. The negative disconfirmation induced by the avatar's lack of behavioral competence should become more salient and detrimental to consumers' cognitive and social responses, resulting in weaker performance outcomes. For example, whereas the realistic, anthropomorphic appearance of HSBC's Amy seems effective because she was programmed to provide basic information to routine customer questions that do not involve high-risk transactions, the very realistic appearance of Nordnet's Amelia could not compensate for her lack of competence to offer stock advice (i.e., high financial risk transactions). Thus, we expect the following moderation effects:

> **Proposition 6:** When an avatar's form realism exceeds its behavioral realism, its positive effect on customers' affective responses is weakened if (1) functional risk is higher, (2) financial risk is higher, and (3) the product is more expensive.
>
> **Proposition 7:** When an avatar's form realism exceeds its behavioral realism, its negative effects on customers' cognitive and social responses are strengthened if (1) functional risk is higher, (2) financial risk is higher, and (3) the product is more expensive.

Conversely, when an avatar's behavioral realism exceeds its form realism, it may have a stronger positive effect on customer performance outcomes under high perceived uncertainty. This is because an avatar's form realism may induce a positive disconfirmation regarding its behavioral competence, which can reassure customers of the usefulness of the information and personalized service provided by the avatar, thereby boosting customers' confidence in their risky purchase decisions. Another pertinent risk in today's world is that of privacy violations (De Haan et al. 2018). If an avatar's behavioral realism exceeds its form realism, the avatar provides reassurance that customers are dealing with an intelligent, nonhuman entity that will not judge them, so it should mitigate social unease and embarrassment. The PTSD therapist Ellie works with highly private matters, involving patients disclosing emotionally and psychologically sensitive information. Such situations might evoke greater concerns if patients cannot tell whether they are dealing with a real human therapist behind the screen.

> **Proposition 8:** When an avatar's behavioral realism exceeds its form realism, its positive effects on customers' cognitive and social responses are stronger if (a) functional risk is higher, (b) financial risk is higher, (c) the product is more expensive, and (d) privacy risk is higher.

As avatars are also increasingly used in mobile apps (e.g., ING's Inge, Progressive's Flo, Bank ABC's Fatema), the choice of mobile or fixed devices as platforms on which the

firm decides to provide an avatar also may be pertinent. Compared with fixed devices (e.g., desktops), mobile devices are particularly entertaining as hedonic information technologies (e.g., video games, MP3 players) have made their way into these portable devices (Xu, Abdinnour, and Chaparro 2017). Moreover, consumers spend more time in online communities when they use mobile rather than fixed devices (Melumad, Inman, and Pham 2019), suggesting that mobile devices are the preferred channel for online social experiences. Thus, avatars that are entertaining and capable of establishing personalized social connections with customers may be especially effective on mobile devices.

> **Proposition 9:** As form realism increases, relative to behavioral realism, the use of mobile devices (compared with fixed devices) strengthens the positive effect of avatars on customers' affective responses.

> **Proposition 10:** As behavioral realism increases, relative to form realism, the use of mobile devices (compared with fixed devices) strengthens the positive effect of avatars on customers' social responses.

Finally, the effects of customers' cognitive, affective, and social responses on performance outcomes may depend on the consumer relationship phase, which refers to the relational trajectory of a customer–seller exchange (Dwyer, Schurr, and Oh 1987; Palmatier et al. 2013). During consumer–avatar interactions, three relationship phases are particularly important: exploration, build-up, and maturity (Jap and Ganesan 2000). During the exploration phase, a consumer is primarily concerned with the potential value and benefits of dealing with the seller, so the perceived informativeness of the avatar becomes especially critical. During this initial stage, entertainment or social engagement provided by the avatar might even detract from the consumer's task objectives and compromise customer outcomes. In this stage, avatar design should allow for behavioral competence, so that the avatar can provide accurate, task-specific, customized information rather than focusing on designing highly attractive or socially engaging avatars. Such an approach should produce a positive disconfirmation for customers' cognitive experience. As the relationship proceeds to the build-up phase, the consumer has experienced some benefits from interactions with the firm's avatars, so socialization processes (e.g., building rapport) become more important. Engaging and fulfilling social experiences can help customers develop long-term commitment to the firm. A positive disconfirmation about the avatar's ability to deepen social bonds might prove especially effective. Finally, during the maturity phase, a consumer satisfied with the cognitive and social benefits of interacting with an avatar may focus less on these factors and instead seek more entertainment value, prioritizing affective responses.

> **Proposition 11:** The effects of customers' cognitive, affective, and social responses on performance outcomes are moderated by the relationship phase, such that (a) the effects of cognitive responses are amplified in the exploration phase but suppressed in the build-up and maturity phases, (b) the effects of affective responses are enhanced in the maturity phase but suppressed in the exploration and build-up phases, and (c) the effects of social responses are strengthened in the build-up phase but weakened in the exploration and maturity phases.

Integrated Framework of Avatar Performance

To synthesize these insights, we offer an integrated framework of avatar performance (Figure 3). This framework provides a visual summary of key insights from our review of extant research and business practices, as well as our development of the avatar taxonomy and propositions. With this framework, we work to advance thought in this emerging, contemporary marketing area by integrating three mediation mechanisms (customers' cognitive, affective, and social responses) together with theory-driven moderators to test theory as well as offer managerial implications.

Managerial Implications and Research Directions for Avatar-Based Marketing

Our integrative analyses of academic research and business practices generate practical implications and research directions for avatar-based marketing, which we group into five key managerially relevant areas: (1) when to deploy avatars, (2) avatar form realism, (3) avatar behavioral realism, (4) form–behavioral realism alignment, and (5) avatar contingency effects (Table 3).

In terms of when and where avatars should be used, avatars seem to be most effective in service-oriented industries (e.g., financial, travel, telecom services), in which the sheer scale of service requirements and customer inquiries can easily overwhelm a company's employees. Avatars can free up employees' time, so that the employees can focus on complex customer needs and offer more value-added services, with greater productivity. Avatars can provide consistent, personalized service and help build emotional connections between the firm and its customers (Corner 2018; Kannan and Bernoff 2019). For companies that serve a large portfolio of customers, avatars can also make it feasible to launch a segmented, multichannel strategy. By offering customers opportunities to engage with avatars through different channels (e.g., social media, company websites, dedicated apps), the firm ensures that it meets each customer's unique needs at the right time and in the right place.

After confirming that an avatar should be implemented, the firm should determine the design of avatar's appearance, with the clear recognition that its form realism is a double-edged sword. On the one hand, a more realistic, anthropomorphic appearance appeals to consumers, because it offers greater entertainment value (Parboteeah, Valacich, and Wells 2009). On the other hand, it elevates customers' expectations of the avatar's behavioral competence, which is much more difficult

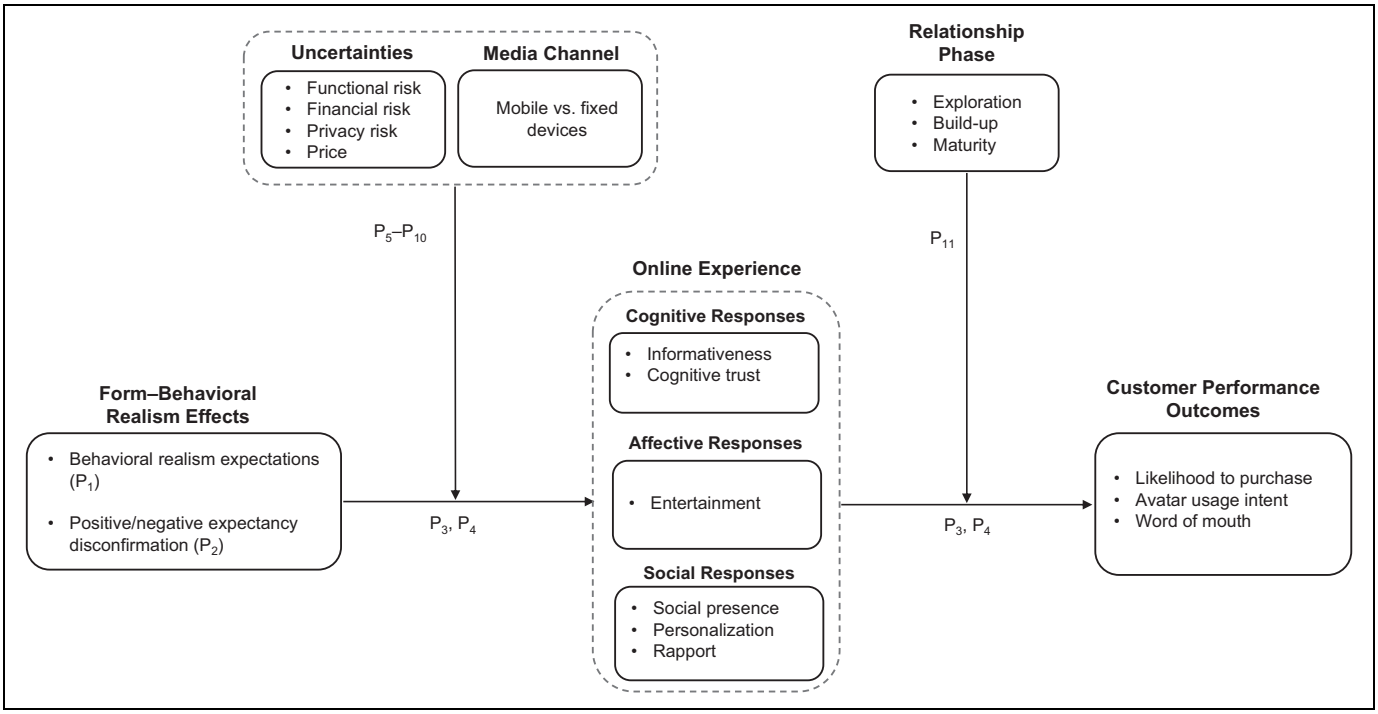


Figure 3. An integrated framework of avatar performance.

and costly to develop. If the avatar's behavioral competence falls short of customers' expectations, they experience a negative disconfirmation, which can decrease their satisfaction. To avoid negative performance outcomes, an avatar's anthropomorphic appearance should not exceed the level of its behavioral competence.

Having decided on the avatar's appearance, companies can design the avatar's behavioral competence. If companies lack the resources to develop high form–high behavioral realism avatars (i.e., digital human avatars), they should allocate more resources to the avatar's behavioral intelligence than to its appearance. A positive disconfirmation of the avatar's cognitive and social competence likely produces an above-average level of customer satisfaction or even customer delight (Finn 2012), which can increase firm performance.

Managers should account for form realism–behavioral realism alignment too. If an avatar has high levels of both form and behavioral realism, customers' high initial expectations about the avatar's behavioral performance will be confirmed. Because customer satisfaction is an additive function of (1) initial expectations of the avatar's behavioral competence and (2) subsequent confirmation or disconfirmation of this expectation (Oliver 1980), the high form realism–high behavioral realism alignment will likely produce high levels of affective, cognitive, and social responses in consumers, as well as better outcomes overall. Additional research is needed to determine the "zone of tolerance" and the precision required when aligning form and behavioral realism of avatars.

Finally, in addition to considering uncertainty factors and media channel choice, design efforts should take the customer relationship phase into account, because the relative effects of customers' cognitive, affective, and social responses differ across relationship stages. For example, during the exploration phase, a positive confirmation regarding an avatar's behavioral realism can ensure good cognitive experiences (e.g., cognitive trust), but during the maturity phase, a more entertaining avatar (e.g., funny, attractive appearance) may prove more effective for sustaining the relationship. Future research that takes a lifecycle approach to avatar design and use could determine avatar effectiveness at each stage and the strategies needed to adapt in accordance with customers' dynamic needs.

Research Directions

Our analysis of avatar design strategies indicates some promising research opportunities. First, propositions derived from our conceptual framework provide opportunities for empirical research, which can be tested using different methods. For example, researchers might collaborate with an avatar design company to manipulate form and behavioral realism in a 2×2 full-factorial experiment, consistent with our taxonomy in Figure 2. These results would provide evidence of the effects of form realism on expectations for behavioral realism (P_1), as well as (dis)confirmations induced by any (mis)alignment (P_2). Moderation tests also might be carried out with lab experiments that allow for manipulations of uncertainty factors (e.g., risk, price, privacy) or channels (e.g., mobile app vs. desktop) (P_5–P_{10}). To test the mediation effects, researchers might use a difference-in-differences field experiment to examine the aggregate main effect. Using a low form realism–low behavioral realism avatar as a baseline (control group), researchers could increase form (treatment 1 group) and behavioral

Table 3. Managerial Implications and Research Directions for Avatar-Based Marketing.

Key Issues/Decisions	Implications	Directions for Future Research
Avatar deployment	• Avatars can be used to humanize a brand with scalable, cost-effective, responsive (24/7), humanlike interactions.	• How can avatar–human collaborations be optimized?
	• Avatars can be used when the scale of service requirements and customer inquiries overwhelm company employees (e.g., financial, travel, telecom services). The employees can use their new free time for more complex issues.	• When the customer encounters a problem with the avatar, which type of "exit ramp" to a human employee is most effective: avatar-initiated, employee-initiated, or customer-initiated, and when during an interaction should it be deployed?
	• Avatars can be used to enhance customer engagement and relationship building through emotional connection, personalization, and service consistency.	• In the event of an avatar service failure, what service recovery strategies should be employed?
	• Avatars can be used to offer multichannel flexibility based on segment preferences (e.g., mobile social media, company website, dedicated apps).	• To what extent can a successful/failed service recovery experience shape customers' (dis)confirmation of the avatar's effectiveness?
Avatar form realism	• Avatars' anthropomorphic appearance is a double-edged sword. A more humanlike appearance appeals to consumers, due to enhanced entertainment value, but it also raises consumers' expectations of the avatar's behavioral competence.	• Which dimension of the avatar's anthropomorphic appearance (i.e., spatial dimension, movement, and other human characteristics) has the greatest impact on consumers' expectations for the avatar's behavioral realism?
	• If behavioral realism falls short of expectations, a negative disconfirmation will be produced, resulting in lower levels of customers' cognitive and social responses to the avatar.	• When might a digital assistant without a visual representation (e.g., Amazon's Alexa) outperform an avatar with an anthropomorphic appearance?
	• Avatars' form realism should not exceed the level of its behavioral competence to avoid unfavorable customer experiences.	• Which avatar form realism elements create the most entertaining avatar experience?
Avatar behavioral realism	• An avatar's behavioral competence is a more impactful design factor than its appearance. In case of a budget constraint, more resources should be allocated to improving avatar behavioral competence than to enhancing its visual appearance.	• What is the role of avatar emotional intelligence, relative to its cognitive abilities, in shaping customers' expectations and overall experience?
	• The higher the avatar's behavioral competence relative to its appearance, the more favorable the customer's cognitive and social experiences, due to a positive disconfirmation.	• What corrective actions can be taken to redress a negative disconfirmation stemming from avatar's behavioral realism?
	• As a caveat, high levels of behavioral competence may produce negative effects (e.g., social desirability bias) when the avatar also has a very realistic anthropomorphic appearance.	• How might other types of avatars (e.g., customers' self-avatars) facilitate social media-based marketing campaigns?
		• Which behavioral realism elements have the greatest impact on customers' expectations?
Form realism–behavioral realism alignment	• High form realism induces high behavioral realism expectations, which will then be confirmed.	• What are the unique benefits, challenges, and risks associated with using high-realism avatars in brand campaigns?
	• The additive nature of high expected behavioral realism and its subsequent confirmation produces satisfactory customer experience with the avatar.	• When would avatar virtual influencers (e.g., Lil Miquela) be more effective than human endorsers in brand promotion?
	• Alignment of high form realism–high behavioral realism results in high levels of customers' affective, cognitive, and social experiences, which subsequently increase firm performance.	• How should hyper-realistic avatars be deployed in marketing campaigns?
Avatar contingency effects	• Consumers' expectations that the avatar's anthropomorphic appearance reflects a comparable	

(continued)

Table 3. (continued)

Key Issues/Decisions	Implications	Directions for Future Research
	level of behavioral competence will be more pronounced when consumers' perceived uncertainty (e.g., the product's functional performance, financial risk) is high.	• How might customer segmentation strategies (e.g., psychographics, benefits sought) inform effective avatar designs?
	• Behavioral realism should account for more weight in avatar design decisions than form realism, especially when customers' perceived uncertainty is high.	• What form realism–behavioral realism elements are most relevant and impactful, given a specific customer segment profile?
	• When the avatar is designed with a low level of behavioral competence, companies should manage customers' expectations by giving the avatar a less realistic, less humanlike appearance (i.e., simplistic avatar). Avoiding a design with high form realism–low behavioral realism (i.e., superficial avatar) becomes even more important when customers' perceived uncertainty is high.	• How do avatar mediation mechanisms differ across customers in different segments?
	• When the exchange entails privacy concerns (e.g., mental health), an avatar characterized by low form realism–high behavioral realism (i.e., intelligent unrealistic avatar) may be more effective than an avatar with high form realism–high behavioral realism (i.e., digital human avatar), because it reassures customers that they will not be judged and promotes more honest responses.	• In which circumstances will avatars likely distract from, rather than contribute to, customers' experience?
	• Use of mobile devices (e.g., smartphones), compared with fixed devices (e.g., desktops), can enhance avatars' impact on consumers' affective and social experiences.	• What contextual factors determine the relative effectiveness of avatars vs. other digital entities (e.g., anthropomorphized products, brand mascots) in online shopping experience?
	• Avatar designs should account for the consumer relationship phase. During the exploration phase, avatar behavioral realism should focus on providing the best cognitive experience; during the build-up phase, avatar design should be directed at enhancing consumers' social experience (e.g., rapport) to promote commitment; during the maturity phase, emphasizing the entertainment value of the avatar (e.g., funny, attractive) may prove more effective in sustaining the established relationship.	

(treatment 2 group) realism. Any significant differences in daily sales across the treatment groups and control group, between the pre- and posttreatment periods, would indicate the external validity of the asymmetric effects of form realism–behavioral realism misalignment (P_3, P_4). To confirm the distinct mediation effects, customers also could be surveyed regarding their cognitive, affective, and social responses, shortly after the treatments, followed by tests of the effects on performance outcomes (e.g., purchase intentions, word of mouth). Alternatively, a lab experiment can be used for differential mediation tests. As for the test of the relationship phase (P_{11}), a cross-sectional survey posted on a retail website that uses an avatar might enable researchers to conduct subgroup analyses or moderated regressions to detect differential effects of mediators across relational phases. A more demanding and rigorous approach would secure panel data from a collaborating retailer that uses avatars and agrees to let the researchers track customers' relational trajectories through longitudinal surveys, while also granting them access to objective customer sales data.

Research opportunities also exist in areas that our framework does not cover. Avatars are designed to enhance the productivity of company employees, rather than replace them altogether, so continued research might investigate how to optimize avatar–human collaborations, especially if problems arise. Various approaches allow customers to switch to a human representative when necessary, using avatar-initiated, employee-initiated, or customer-initiated "exit ramps." These approaches can differ in their nature (proactive vs. reactive) and timing (early vs. late in the interaction). Thus, determining

how and *when* human intervention is introduced could provide significant insights for achieving service recovery and ensuring customers' overall satisfaction with and commitment to the firm. In this article, we have highlighted the influence of an avatar's form realism on the overall customer experience. Additional research should uncover which specific dimensions of the avatar's anthropomorphic appearance exert the strongest impacts on customers' behavioral realism expectations, which form realism elements create the most entertaining avatar experience, or when a digital assistant without an anthropomorphic appearance (e.g., Amazon's Alexa) would perform better than an avatar. For example, if the avatar's form realism exceeds its behavioral realism, which subset of behavioral elements is most likely to lead to customers' expectation disconfirmation? What corrective actions would be effective in addressing negative disconfirmations? Research also could delve deeper into the effects of avatars' emotional intelligence, relative to their cognitive intelligence, in shaping customers' expectancy (dis)confirmations and overall experience.

Another type of avatar that is growing in popularity is customers' self-avatars, created with virtual model technology (Smith, Johnston, and Howard 2011). To inform research into these applications, the typology we developed would need to be modified to reflect the unique characteristics of self-avatars (e.g., resemblance to self). Some designer brands (e.g., Gucci) have successfully engaged customers to dress their self-avatars in branded products, then share them on social media (Carson 2019). The limited research on self-avatars has focused primarily on enclosed online environments (e.g., retailer's website), not social media (Cho and Schwarz 2012; Fiore, Kim, and Lee 2005), indicating the pressing need for insights into *how*, *why*, and *when* self-avatars could perform in social media marketing campaigns.

Brands also have turned to virtual influencers (3D, computer-generated personalities) instead of or in addition to human influencers for online marketing campaigns. Powered by advanced AI, these avatars can attract significant followers on social media platforms. For example, with almost 3 million Instagram followers, Lil Miquela has endorsed brands such as Prada and Calvin Klein (Abad 2019; Bezamat 2018). This avatar actively replies to social media comments, appears in publications like *Vogue*, and even participates in live media interviews (Chichioco 2019). While virtual influencers offer unique benefits to brands such as content control and versatility, they also create potential risks. For example, 61% of consumers assert that authentic, relatable content is the primary appeal of human influencers (Penny 2019), but only 15% of followers of virtual influencers describe them as credible (Chowdhary 2019). Moreover, because the virtual influencer avatar is not a human, the brand it endorses ultimately is held responsible for its actions. Academic research has yet to investigate the unique benefits, risks, and operational mechanisms associated with avatar-based virtual influencer marketing.

Future research might also explore avatar-based targeting strategies. Demographics, psychographics, and benefits sought are widely used customer segmentation bases (Tynan and Drayton 1987); they also might be used to predict which customers will be best served by a given type of avatar. For example, customer's demographic traits might interact with the avatar's demographic attributes or behavioral elements to influence customer's cognitive, affective, and social responses to the online experience. Creating a personality or decision-making style for an avatar that matches those of the customer might be an effective, psychographics-based design strategy (Al-Natour, Benbasat, and Cenfetelli 2011). Segmenting markets on the basis of in-depth analyses of the motives that lead certain people to interact with avatars could also inform benefits-based avatar deployment strategies (Brandtzaeg and Følstad 2018). Distinct mediation mechanisms could be uncovered across different customer segments to inform idiosyncratic avatar designs. Research is also needed to determine when avatars may distract from, rather than contribute to, the customer's experience, as well as find strategies to address these challenges. Finally, to extend beyond our focus on the relative effects of the different types of avatars established by our taxonomy, future research might compare the effects of avatars with the impacts of other digital representations such as emoji, anthropomorphized products, brand mascots, or voice-only digital assistants. Understanding when and how avatars, versus these alternative marketing tools, are more effective in influencing online shopping experience and performance outcomes demands further scientific inquiry.

Conclusion

Rapid increases in the use of avatars have been fueled by two main factors: advances in digital technologies and increasing reliance on online experiences among both consumers and firms. The use of avatars is projected to grow by 35% annually (Globe Newswire 2019). However, the effectiveness of avatars continues to be uncertain, so we offer an integrated theoretical framework to establish definitional and conceptual clarity, synthetize academic research and business practices, and offer insights and propositions that provide managerial implications and an agenda for future research. The proposed definition of avatars, as digital entities with anthropomorphic appearance, controlled by a human or software, with an ability to interact, helps us establish a design typology, which in turn gives academics and managers insights into how to isolate elements that make avatars more or less effective for specific goals.

We synthesize academic literature and business practices by offering a 2×2 form realism–behavioral realism taxonomy, which in turn enables us to derive propositions regarding the effectiveness of avatars in marketing. The level of alignment between an avatar's form realism and behavioral realism, according to several contingencies, can provide a parsimonious explanation of when an avatar will be most effective. With insights gained from our investigation of fundamental avatar elements, extant research, and business practices, we develop an integrative framework of avatar performance that offers theoretical insights, research propositions, managerial implications, and future research directions.

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